

MTH1000 Class Test 18 November 2025

Attempt all questions. Submit your results on Loop, either on an R file or an R file and Word. Use R functions for calculations where possible. The time allowed is 1 hour and 50 minutes. Marks are shown in brackets.

1. The values of μ_x for the AM92 mortality table for ages in the range $17 \leq x \leq 119$ are calculated using the following formula:

$$\mu_x = a_0 + a_1 t + \exp[b_0 + b_1 t + b_2(2t^2 - 1)]$$

where $t = \frac{x-70}{50}$ and $a_0 = 0.00005887$, $a_1 = -0.0004988$, $b_0 = -4.363378$, $b_1 = 5.544956$ and $b_2 = -0.620345$.

- (i) Construct a function in R to calculate μ_x for an inputted age x . [5]
- (ii) You are given that q_x can be approximated using the relationship

$$q_x = \exp\left(1 - \mu_{x+\frac{1}{2}}\right)$$

Calculate an approximate value for the curtate expectation of life e_{35} , using this relationship for q_x . [10]

[Total 15]

2. Let T_{40} be the random variable denoting the expected future lifetime of lives currently aged 40. It is believed that T_{40} follows the $W(0.00002, 3.6)$ distribution.

- (i) (a) Generate a sample of 10,000 values of T_{40} , setting a seed of 100 and storing the simulated values in the vector *sims*.
- (b) Estimate $P(T_{40} > 10)$ using your sample. [8]

Under this model, the hazard at age $40 + t$ is given by:

$$\mu_{40+t} = \alpha \beta t^{\beta-1}$$

where $\alpha = 0.00002$ and $\beta = 3.6$.

- (ii) (a) Construct a function R to calculate μ_x , for an inputted age, $x \geq 40$.
- (b) Calculate $P(T_{50} > 10)$ using your function for μ_x . [5]
- (iii) Comment on you answers to parts (i)(b) and (ii)(b). [2]

[Total 15]

3. The table below shows the number of circuits a group of 10 army recruits were able to complete in a training exercise and their ages, split by sex.

Complete circuits	Males	Females
1		
2		30
3	21,20	24,25
4		19
5	18,18,17	
6		21

Two Cox proportional hazards models are being considered for modelling the result :

Model 1: $\ln\lambda(t, z_1) = \ln\lambda_0(t) + \beta_1 z_1$

Model 2: $\ln\lambda(t, z_1, z_2) = \ln\lambda_0(t) + \beta_1 z_1 + \beta_2 z_2$

Here:

- $\lambda(t, z_1)$ and $\lambda(t, z_1, z_2)$ are the hazard rates for ‘dropping out’, i.e. for the recruit reaching their limit in terms of the number of circuits they can complete
- z_1 is the recruit’s sex ($z_1 = 0$ for males, $z_1 = 1$ for females)
- z_2 is the recruit’s age.

- (i) Construct a data frame that contains a row for each recruit and the following columns:
 - Age – the age of the recruit
 - Sex – the sex of the recruit
 - Circuits – the number of circuits completed by the recruit
 - Status – whether each recruit experienced the death event (1) or was censored (0)

[6]
 - (ii)
 - (a) Fit Model 1 to the data.
 - (b) State the estimated value of β_1 and comment on this value. [10]
 - (iii)
 - (a) Fit Model 2 to the data.
 - (b) State the estimated values of β_1 and β_2 and comment on these values. [10]
 - (iv) Determine if Model 2 is a significant improvement over Model 1. [6]
- [Total 32]

4. Before answering this question, the 'markovchain' R package should be loaded into R using the following code:

```
install.packages("markovchain")  
library(markovchain)
```

A three-state Markov chain model consisting of healthy ('H'), sick ('S'), and dead ('D') states has the following weekly transition probabilities:

$$\begin{aligned}p_{HH} &= 0.97 \\p_{HS} &= 0.029 \\p_{HD} &= 0.001 \\p_{SS} &= 0.7 \\p_{SH} &= 0.25 \\p_{SD} &= 0.05\end{aligned}$$

- (i) Construct a 'markovchain' object for the transition matrix for the above Markov chain model. [2]
- (ii) Calculate the probability that, given a life is healthy now, the life will be sick at time 3 weeks. [3]
- (iii) Calculate the probability that, given a life is healthy now, the life will be sick at some point in the next 52 weeks. [3]
- (iv) Calculate the probability that, given a life is healthy now, the life will remain healthy for the entire year. [1]

A new, disease has become prevalent, and it is proposed that the HSD model above be amended to model the effects of this disease. Lives become ill for a period of time and then either recover or die. Once recovered, a life is deemed immune and cannot become ill again.

It has been decided to use a Markov jump model instead of a chain model. The Markov model consists of four states: healthy ('H'), sick ('S'), recovered ('R') and dead ('D') state. The following daily transition rates have been estimated:

$$\begin{aligned}\mu_{SD} &= 5\% \\ \mu_{SR} &= 14\%\end{aligned}$$

The rate from healthy to infected, μ_{HS} , is equal to b multiplied by i_t where b is a constant and i_t is the proportion of lives in the sick state at time t . All other transition rates are zero.

- (v) Explain why it may be preferable to set $\mu_{HS} = bi_t$ (as above), rather than using a constant value (as has been done for the other rates). [1]

- (vi) Give an example of a scenario that would be expected to result in a particularly high value of b . [2]

The proposed value of b is 0.35, and it is assumed that at time 0, 1% of the population are in the sick state and the remaining 99% are in the healthy state.

- (vii) Using this revised model and a step length of 0.01 days, calculate the occupancy probabilities in each of the four states from $t = 0.01$ to $t = 100$ days inclusive. You should output your answers for each value of t to successive rows of a matrix. [12]

- (viii) Plot a graph showing the probabilities in part (vii) with suitably labelled axes and making clear which plot components correspond to which state. [5]

- (ix) Calculate the probability that a life that is healthy at time 0 is sick after:

(a) 6 days.

(b) 25 days. [4]

- (x) Calculate the expected present value of a daily rate of €1 payable while sick using a force of interest of 5% p.a. [5]

[Total 38]